

Miniature life

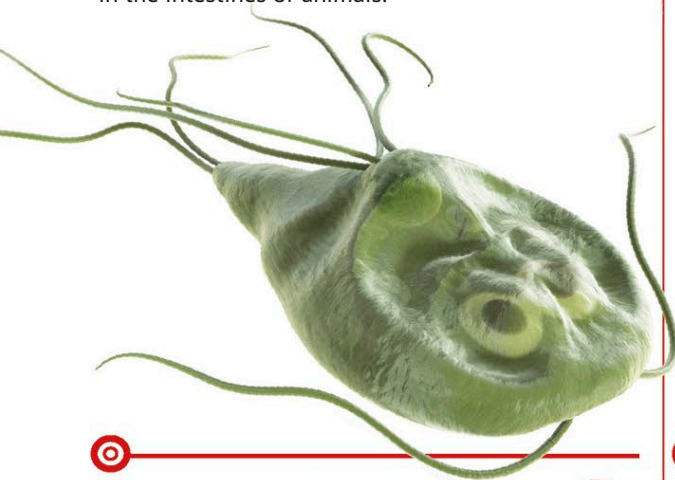
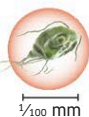
Some organisms are so tiny that thousands of them can live out their lives in a single drop of water.

The minuscule home of the microbe, or microorganism, is a place where sand grains are like giant boulders and the slightest breeze feels like a hurricane. These living things can only be seen through a microscope, but manage to find everything they need to thrive in soil, oceans, or even deep inside the bodies of bigger animals.

GIARDIA

Kingdom: Protozoa

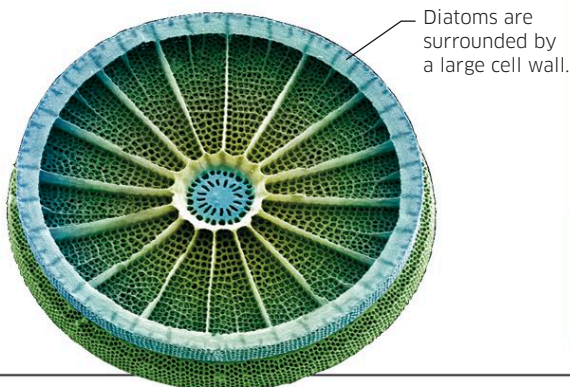
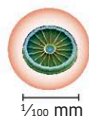
Animal-like microbes that are single-celled are called protozoans. Some, such as amoebas, use extensions of cytoplasm (cell material) to creep along. Others, such as giardia, swim, and absorb their food by living in the intestines of animals.



DIATOM

Kingdom: Algae

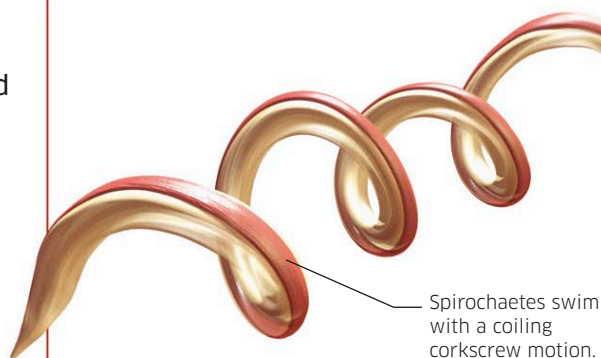
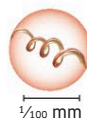
The biggest algae grow as giant seaweeds, but many, such as diatoms, are microscopic single cells. All make food by photosynthesis, forming the bottom of many underwater food chains that support countless lives.



SPIROCHAETE

Kingdom: Bacteria

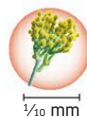
Any place good for life can be home to bacteria—the most abundant kinds of microorganisms on the planet. They are vital for recycling nutrients, although some—such as the corkscrew-shaped spirochaetes—are parasites that cause disease in humans and other animals.



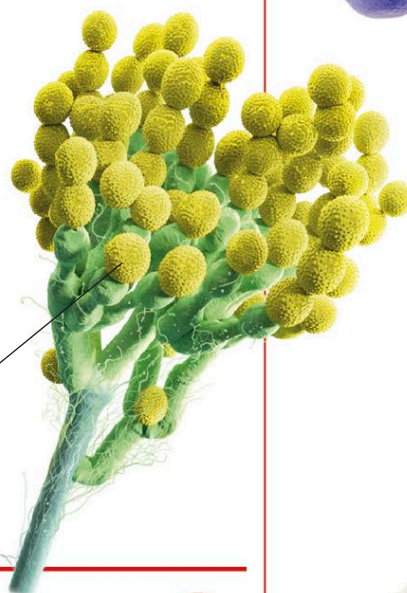
PENICILLIUM

Kingdom: Fungi

The microscopic filaments of fungi smother dead material, such as leaf litter, so their digestive juices can break it down. When their food runs out, they scatter dustlike spores, which grow into new fungi.



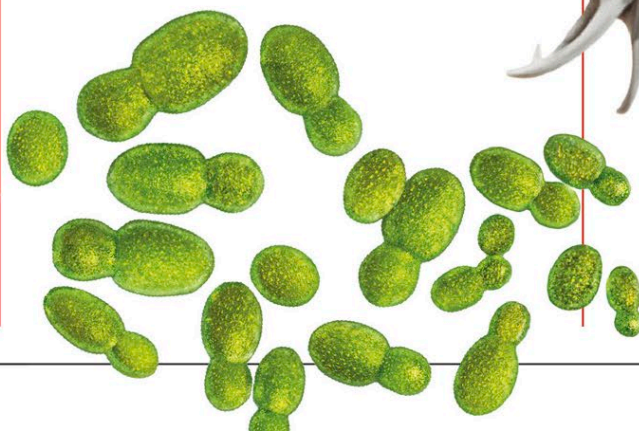
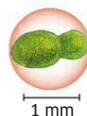
Single-celled spherical spores grow from the *Penicillium* fungus before detaching.



WATERMEAL

Kingdom: Plants

The smallest plant, called watermeal, floats on ponds, blanketing the surface in its millions. A hundred could sit comfortably on a fingertip, each one carrying a tiny flower that allows it to reproduce.



THERMOPLASMA

Kingdom: Archaea

These microbes look like bacteria, but are a distinct life form. Many, like the *Thermoplasma volcanium*, live in the most hostile habitats imaginable, such as hot pools of concentrated acid.

